SBT-DF204 — Computer Forensics Case Study

Case Study 1: Investigating Harassment Email Traffic With Wireshark

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Course Title : Computer Forensics Case Study

Lab Title: Investigating Harassment Email Traffic With Wireshark Forensics

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Executive Summary

This investigation examined network traffic related to a reported harassment email. The analysis of the PCAP file identified a client computer that communicated with the web service used to send the message. The traffic included an HTTP POST request containing information that matched the reported harassment message.

The investigation also linked the relevant network activity to a specific MAC address and found an email-related cookie associated with the same device. The available evidence was compared with the Chemistry 109 roster to assess the possible identity of the user.

The evidence shows a strong connection between the device, the web activity, and the reported message.

Objective

The objective of this investigation is to examine the network traffic in the Nitroba PCAP file to identify evidence related to the reported harassment email. The investigation aims to identify the client computer,

examine the web request used to send the message, link the activity to the relevant device, and assess the available evidence while maintaining the integrity of the original evidence.

Tools Used

The following tools were used during the investigation:

*Wireshark** – Used to open and examine the PCAP file and inspect network packets, HTTP traffic, DNS queries, TCP streams, and Ethernet information.
*TShark** – Used from the Kali Linux terminal to filter and extract specific packet information from the capture.
*Kali Linux** – Used as the forensic analysis environment.
*SHA-256** – Used to verify the integrity of the original PCAP file and the working copy.
*Linux terminal commands** – Used for downloading, copying, checking file size, calculating hashes, and preserving the evidence.
*Digital Corpora** – Used as the source of the Nitroba PCAP and for comparing the published file checksum.

Methodology

The investigation started by acquiring the Nitroba PCAP file and checking its integrity using SHA-256 hashing. A working copy was created so that the original evidence could be preserved.

The working copy was then opened in Wireshark for analysis. Different filters were used to identify the relevant DNS queries, HTTP requests, client and server IP addresses, TCP streams, MAC addresses, and cookies. The HTTP POST request was also examined to identify the information submitted through the web service.

TShark was used to extract and confirm specific packet information from the capture. The evidence was reviewed using packet numbers, timestamps, IP addresses, MAC addresses, and other network details. The timestamps were treated as UTC when reviewing the packet timeline. The findings were then compared with the available scenario information and Chemistry 109 roster to assess the possible connection between the device and the reported activity.

```
┌──(kali㉿kali)-[~]
└─$ mkdir -p ~/SBT-DF204-Case1

┌──(kali㉿kali)-[~]
└─$ cd ~/SBT-DF204-Case1

┌──(kali㉿kali)-[~/SBT-DF204-Case1]
└─$ mkdir -p ~/Nitroba-Forensics
cd ~/Nitroba-Forensics

┌──(kali㉿kali)-[~/Nitroba-Forensics]
└─$ pwd
/home/kali/Nitroba-Forensics
```

```
┌──(kali㉿kali)-[~/Nitroba-Forensics]
└─$ wget -O nitroba.pcap.part https://digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitr
oba/nitroba.pcap && mv nitroba.pcap.part nitroba.pcap
ls -l nitroba.pcap
--2026-10-06 19:15:49--  https://digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/n
itroba.pcap
Resolving digitalcorpora.s3.amazonaws.com (digitalcorpora.s3.amazonaws.com)... 16.15.35.14, 52.92
.165.33, 16.15.39.103, ...
Connecting to digitalcorpora.s3.amazonaws.com (digitalcorpora.s3.amazonaws.com)|16.15.35.14|:443.
.. connected.
HTTP request sent, awaiting response... 200 OK
Length: 56180821 (54M) [binary/octet-stream]
Saving to: 'nitroba.pcap.part'

nitroba.pcap.part        100%[===================>]  53.58M  5.51MB/s    in 10s

2026-10-06 19:16:01 (5.21 MB/s) - 'nitroba.pcap.part' saved [56180821/56180821]

-rw-rw-r-- 1 kali kali 56180821 Nov 21  2020 nitroba.pcap
```

```
┌──(kali㉿kali)-[~/Nitroba-Forensics]
└─$ sha256sum nitroba.pcap
2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb  nitroba.pcap
```

```
┌──(kali㉿kali)-[~/Nitroba-Forensics]
└─$ cd ~/Nitroba-Forensics
md5sum nitroba.pcap nitroba_working.pcap
sha1sum nitroba.pcap nitroba_working.pcap
sha256sum nitroba.pcap nitroba_working.pcap
9981827f11968773ff815e39f5458ec8  nitroba.pcap
9981827f11968773ff815e39f5458ec8  nitroba_working.pcap
65656392412add15f93f8585197a8998aaeb50a1  nitroba.pcap
65656392412add15f93f8585197a8998aaeb50a1  nitroba_working.pcap
2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb  nitroba.pcap
2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb  nitroba_working.pcap
```

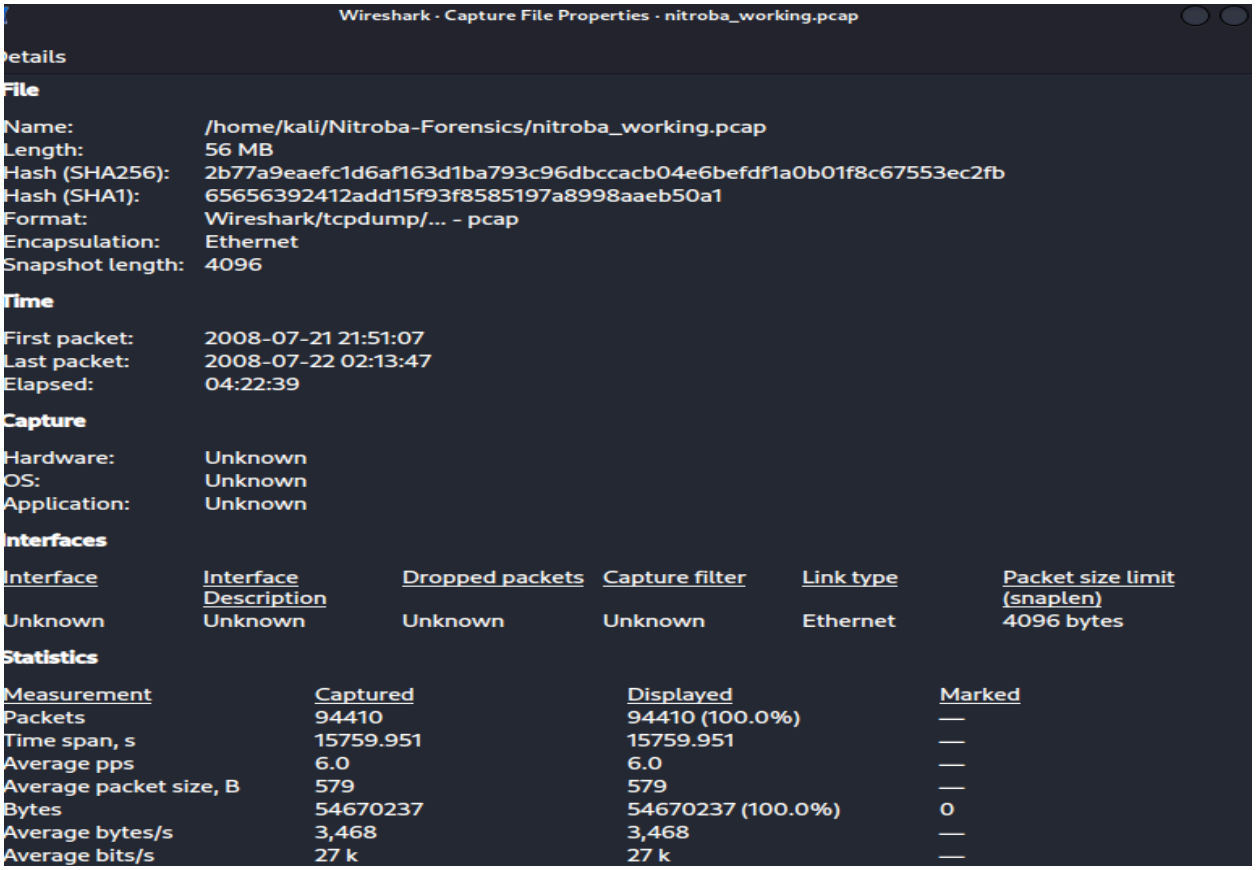
Acquire and preserve the evidence

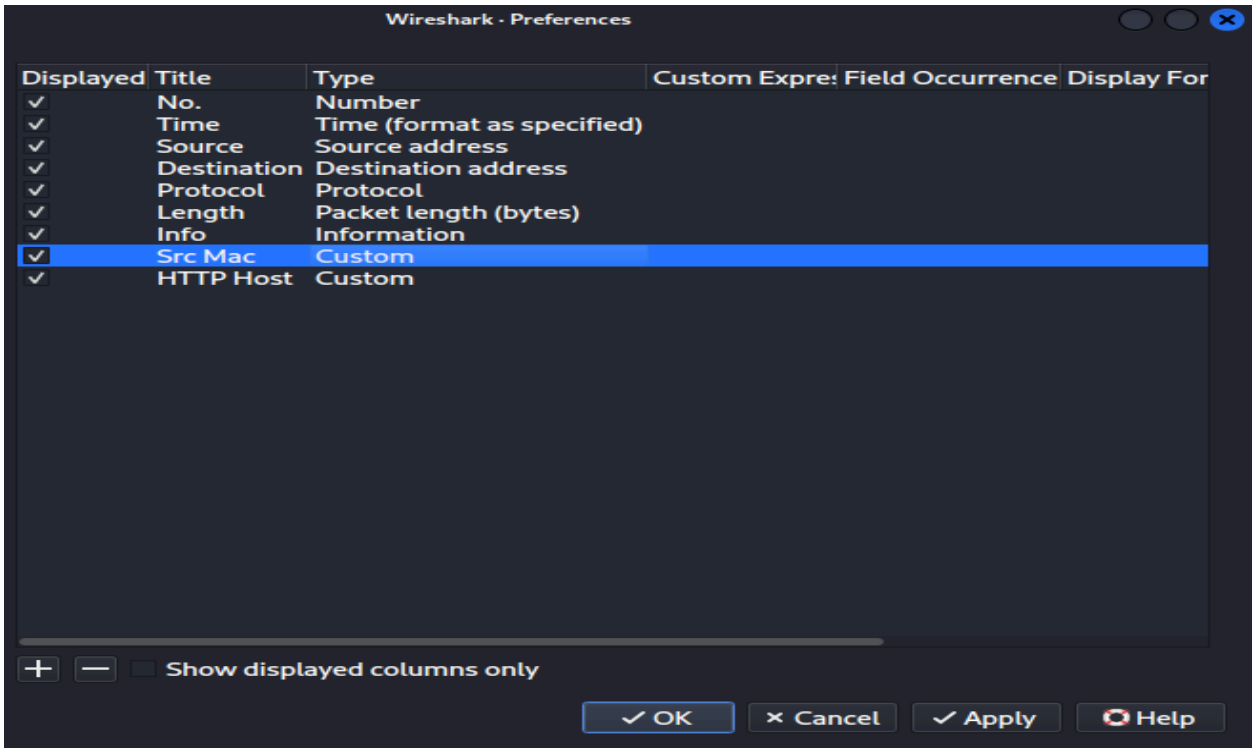
```
┌──(kali㉿kali)-[~/Nitroba-Forensics]
└─$ cd ~/Nitroba-Forensics
chmod 444 nitroba.pcap
cp nitroba.pcap nitroba_working.pcap
{
  echo "Evidence File: nitroba.pcap"
  echo "Source: https://digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pc
ap"
  echo "Download (local EDT): 2026-10-06 19:15:49 to 19:16:01"
  echo "Download (UTC): 2026-10-06 23:15:49 to 23:16:01"
  echo "Working Copy: nitroba_working.pcap"
  echo "Record created (local):"; date
  echo "Record created (UTC):"; date -u
  echo; echo "File Size:"
  stat -c '%n: %s bytes' nitroba.pcap nitroba_working.pcap
  echo; echo "Permissions:"
  ls -l nitroba.pcap nitroba_working.pcap
  echo; echo "SHA-256:"
  sha256sum nitroba.pcap nitroba_working.pcap
} | tee evidence_preservation.txt
Evidence File: nitroba.pcap
Source: https://digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pcap
Download (local EDT): 2026-10-06 19:15:49 to 19:16:01
Download (UTC): 2026-10-06 23:15:49 to 23:16:01
Working Copy: nitroba_working.pcap
Record created (local):
Wed Oct  7 10:11:45 AM EDT 2026
Record created (UTC):
Wed Oct  7 02:11:45 PM UTC 2026

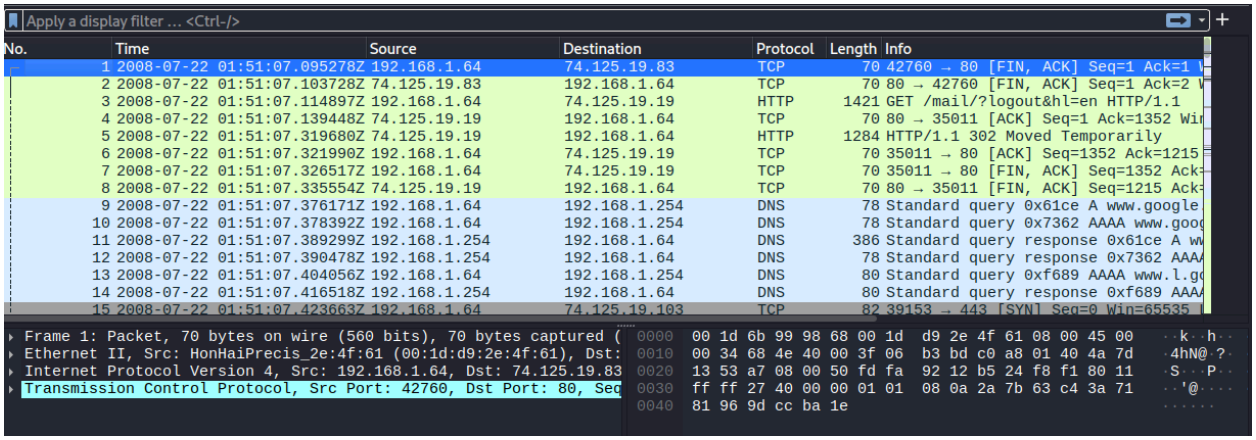
File Size:
nitroba.pcap: 56180821 bytes
nitroba_working.pcap: 56180821 bytes

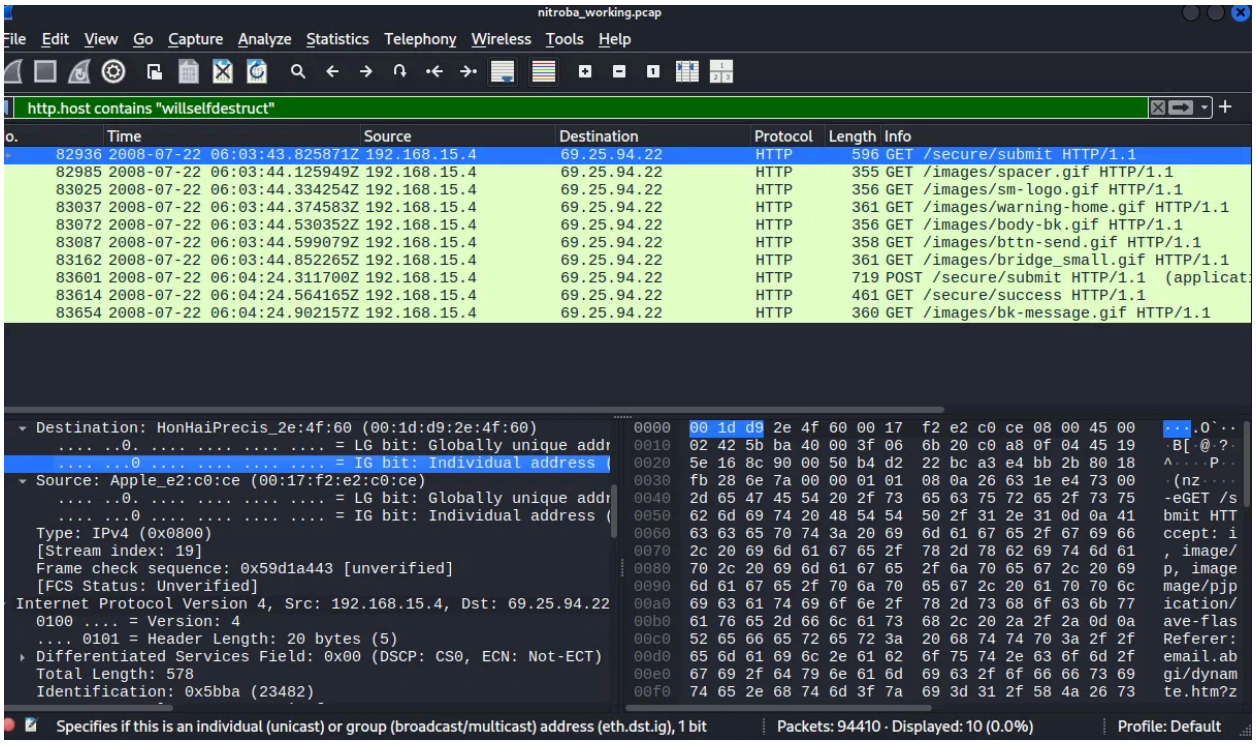
Permissions:
-r--r--r-- 1 kali kali 56180821 Nov 21  2020 nitroba.pcap
-r--r--r-- 1 kali kali 56180821 Oct  7 10:11 nitroba_working.pcap

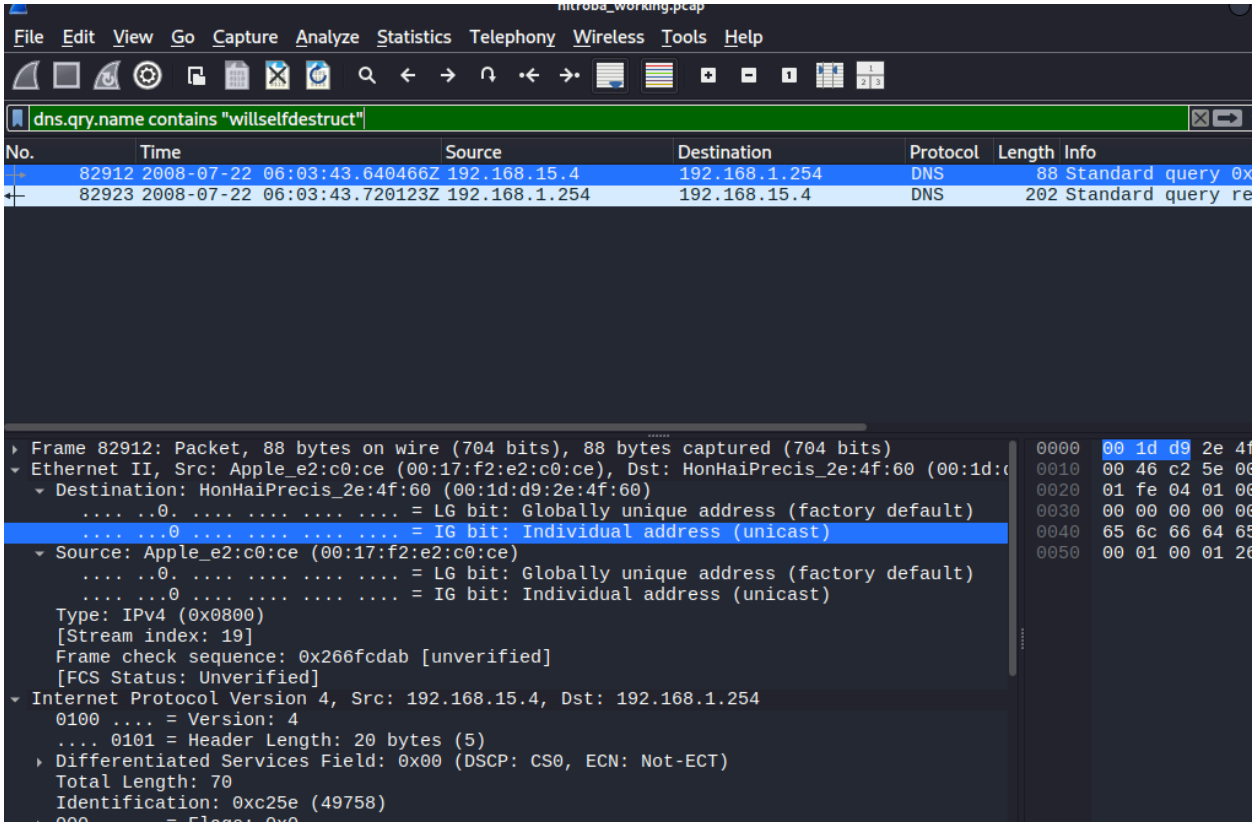
SHA-256:
2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb  nitroba.pcap
2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb  nitroba_working.pcap
```

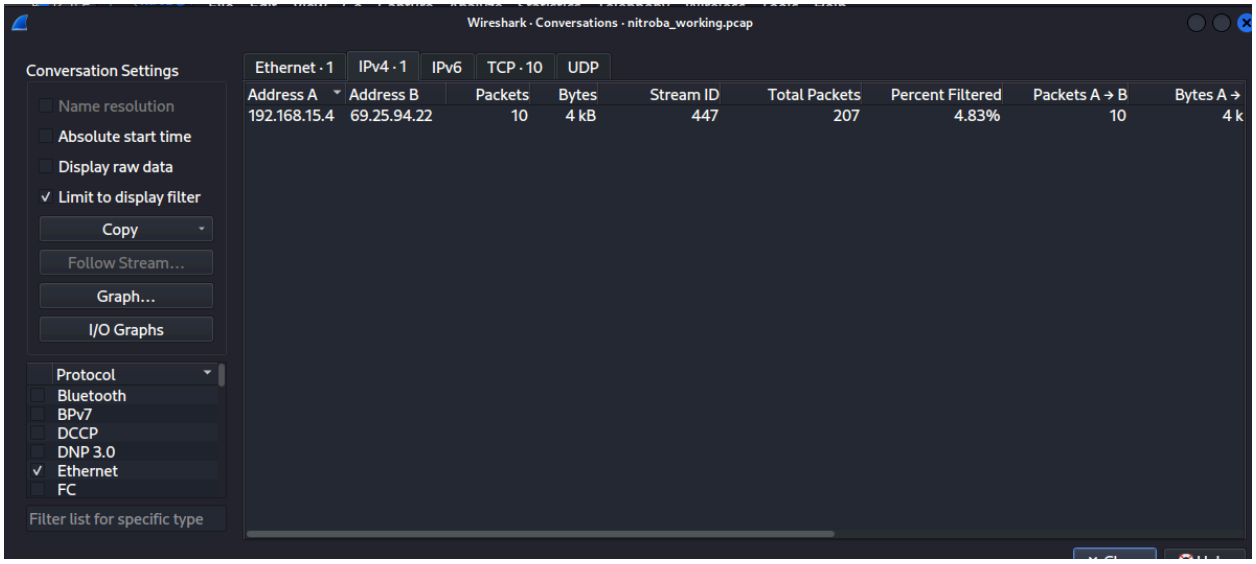

Wireshark · Capture File Properties · nitroba_working.pcap
Details
File
Name: /home/kali/Nitroba-Forensics/nitroba_working.pcap
Length: 56 MB
Hash (SHA256): 2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb
Hash (SHA1): 65656392412add15f93f8585197a8998aaeb50a1
Format: Wireshark/tcpdump/... - pcap
Encapsulation: Ethernet
Snapshot length: 4096
Time
First packet: 2008-07-21 21:51:07
Last packet: 2008-07-22 02:13:47
Elapsed: 04:22:39
Capture
Hardware: Unknown
OS: Unknown
Application: Unknown
Interfaces
Interface | Interface Description | Dropped packets | Capture filter | Link type | Packet size limit (snaplen)
Unknown | Unknown | Unknown | Unknown | Ethernet | 4096 bytes
Statistics
Measurement | Captured | Displayed | Marked
Packets | 94410 | 94410 (100.0%) | —
Time span, s | 15759.951 | 15759.951 | —
Average pps | 6.0 | 6.0 | —
Average packet size, B | 579 | 579 | —
Bytes | 54670237 | 54670237 (100.0%) | 0
Average bytes/s | 3,468 | 3,468 | —
Average bits/s | 27 k | 27 k | —

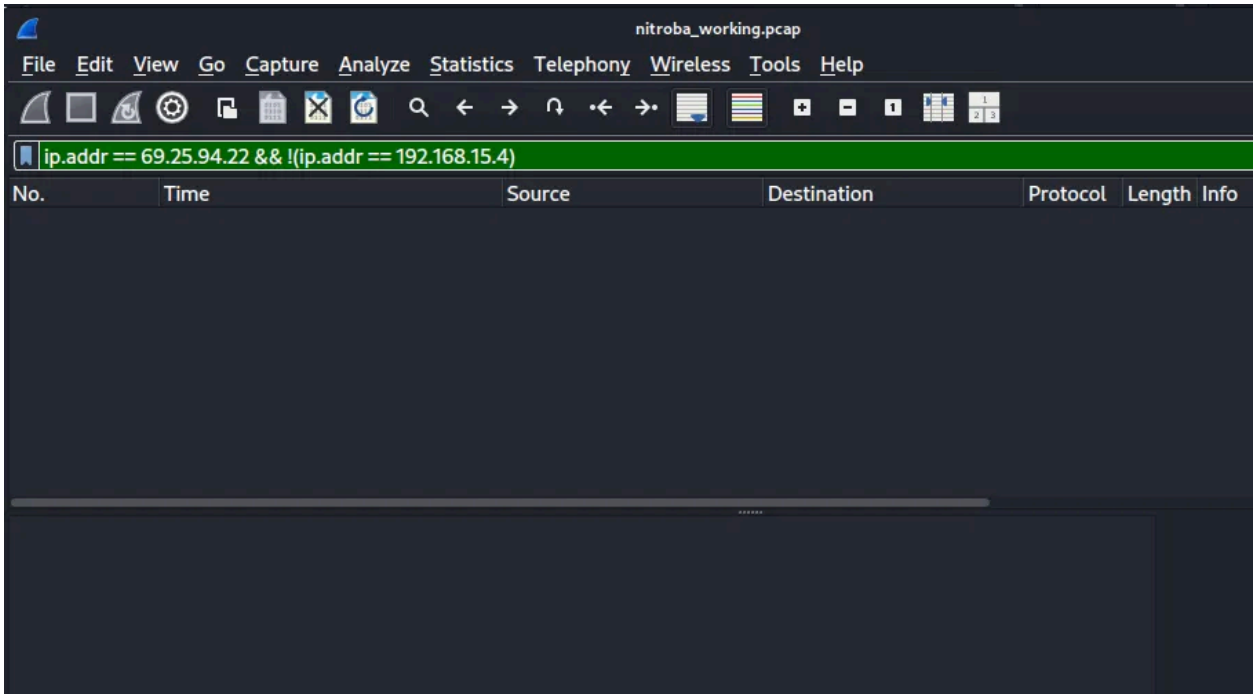

Wireshark · Preferences
Displayed | Title | Type | Custom Expre | Field Occurrence | Display For
No. | Number
Time | Time (format as specified)
Source | Source address
Destination | Destination address
Protocol | Protocol
Length | Packet length (bytes)
Info | Information
Src Mac | Custom
HTTP Host | Custom
Show displayed columns only
OK
Cancel
Apply
Help

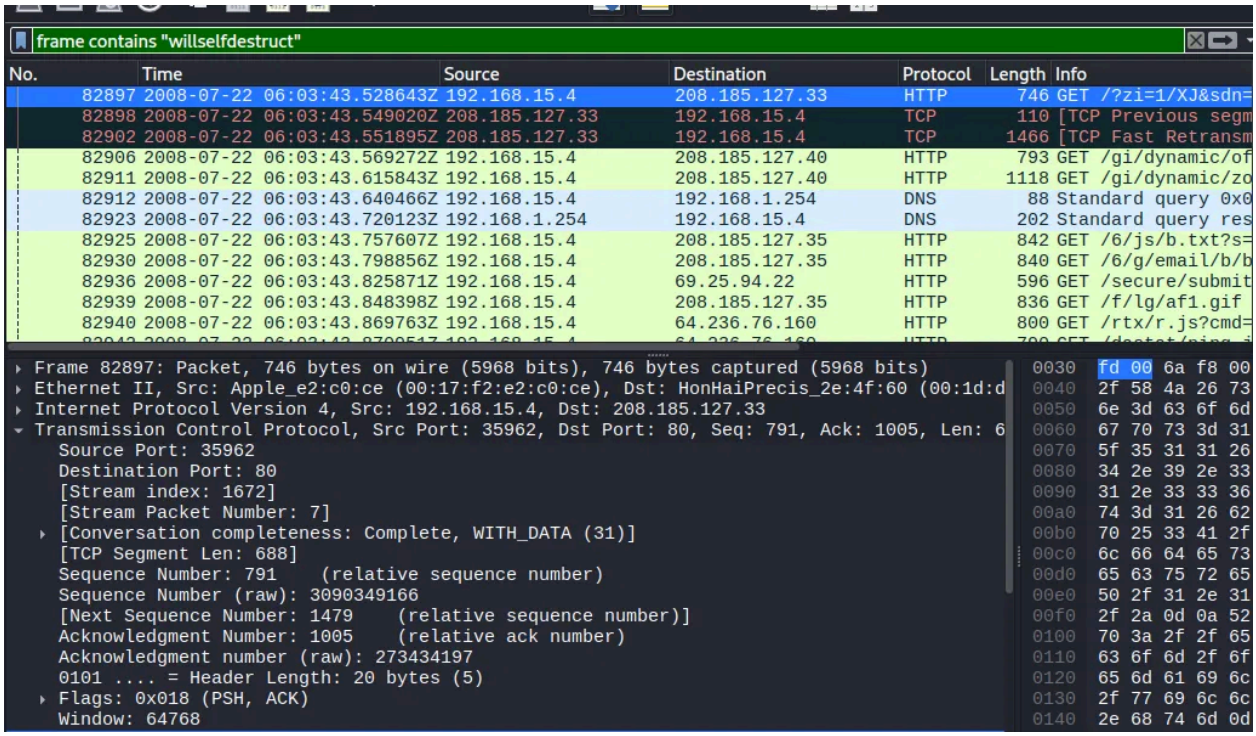

Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
1 2008-07-22 01:51:07.095278Z 192.168.1.64 74.125.19.83 TCP 70 42760 → 80 [FIN, ACK] Seq=1 Ack=1
2 2008-07-22 01:51:07.103728Z 74.125.19.83 192.168.1.64 TCP 70 80 → 42760 [FIN, ACK] Seq=1 Ack=2
3 2008-07-22 01:51:07.114897Z 192.168.1.64 74.125.19.19 HTTP 1421 GET /mail/?logout&hl=en HTTP/1.1
4 2008-07-22 01:51:07.139448Z 74.125.19.19 192.168.1.64 TCP 70 80 → 35011 [ACK] Seq=1 Ack=1352
5 2008-07-22 01:51:07.319680Z 74.125.19.19 192.168.1.64 HTTP 1284 HTTP/1.1 302 Moved Temporarily
6 2008-07-22 01:51:07.321990Z 192.168.1.64 74.125.19.19 TCP 70 35011 → 80 [ACK] Seq=1352 Ack=1215
7 2008-07-22 01:51:07.326517Z 192.168.1.64 74.125.19.19 TCP 70 35011 → 80 [FIN, ACK] Seq=1352 Ack=
8 2008-07-22 01:51:07.335554Z 74.125.19.19 192.168.1.64 TCP 70 80 → 35011 [FIN, ACK] Seq=1215 Ack=
9 2008-07-22 01:51:07.376171Z 192.168.1.64 192.168.1.254 DNS 78 Standard query 0x61ce A www.google
10 2008-07-22 01:51:07.378392Z 192.168.1.64 192.168.1.254 DNS 78 Standard query 0x7362 AAAA www.goo
11 2008-07-22 01:51:07.389299Z 192.168.1.254 192.168.1.64 DNS 386 Standard query response 0x61ce A ww
12 2008-07-22 01:51:07.390478Z 192.168.1.254 192.168.1.64 DNS 78 Standard query response 0x7362 AAA
13 2008-07-22 01:51:07.404056Z 192.168.1.64 192.168.1.254 DNS 80 Standard query 0xf689 AAAA www.l.g
14 2008-07-22 01:51:07.416518Z 192.168.1.254 192.168.1.64 DNS 80 Standard query response 0xf689 AAA
15 2008-07-22 01:51:07.423663Z 192.168.1.64 74.125.19.103 TCP 82 39153 → 443 [SYN] Seq=0 Win=65535
Frame 1: Packet, 70 bytes on wire (560 bits), 70 bytes captured (
Ethernet II, Src: HonHaiPrecis_2e:4f:61 (00:1d:d9:2e:4f:61), Dst:
Internet Protocol Version 4, Src: 192.168.1.64, Dst: 74.125.19.83
Transmission Control Protocol, Src Port: 42760, Dst Port: 80, Seq
0000 00 1d 6b 99 98 68 00 1d d9 2e 4f 61 08 00 45 00
0010 00 34 68 4e 40 00 3f 06 b3 bd c0 a8 01 40 4a 7d
0020 13 53 a7 08 00 50 fd fa 92 12 b5 24 f8 f1 80 11
0030 ff ff 27 40 00 00 01 01 08 0a 2a 7b 63 c4 3a 71
0040 81 96 9d cc ba 1e

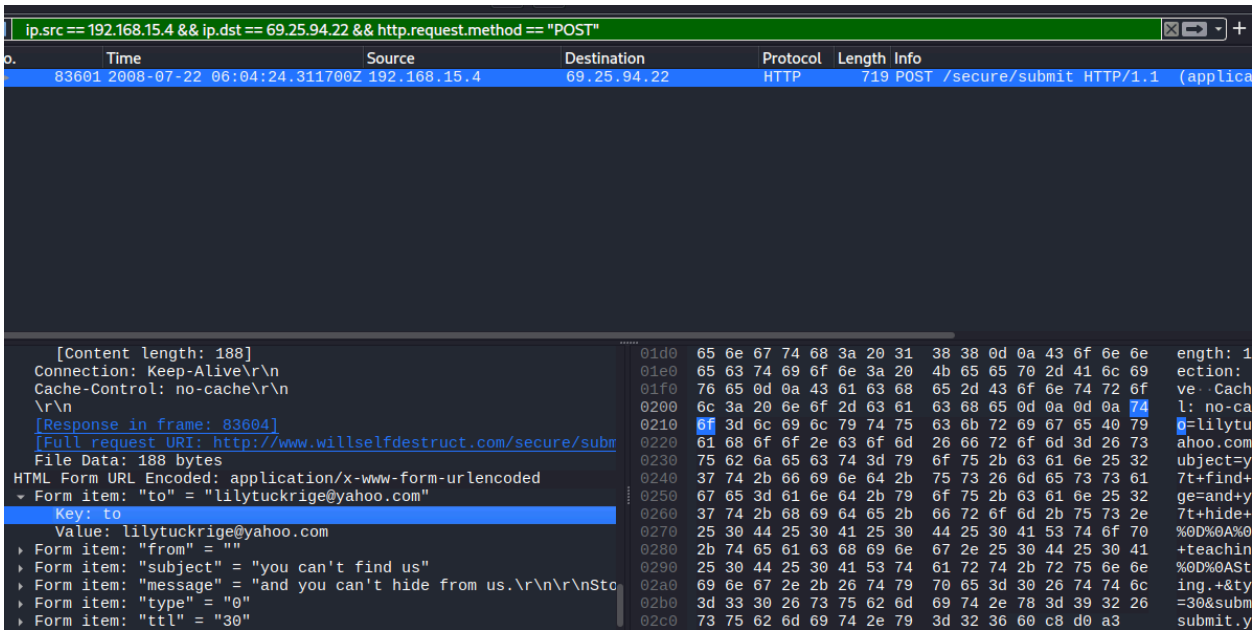

nitroba_working.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
http.host contains "willselfdestruct"
o. Time Source Destination Protocol Length Info
82936 2008-07-22 06:03:43.825871Z 192.168.15.4 69.25.94.22 HTTP 596 GET /secure/submit HTTP/1.1
82985 2008-07-22 06:03:44.125949Z 192.168.15.4 69.25.94.22 HTTP 355 GET /images/spacer.gif HTTP/1.1
83025 2008-07-22 06:03:44.334254Z 192.168.15.4 69.25.94.22 HTTP 356 GET /images/sm-logo.gif HTTP/1.1
83037 2008-07-22 06:03:44.374583Z 192.168.15.4 69.25.94.22 HTTP 361 GET /images/warning-home.gif HTTP/1.1
83072 2008-07-22 06:03:44.530352Z 192.168.15.4 69.25.94.22 HTTP 356 GET /images/body-bk.gif HTTP/1.1
83087 2008-07-22 06:03:44.599079Z 192.168.15.4 69.25.94.22 HTTP 358 GET /images/bttn-send.gif HTTP/1.1
83162 2008-07-22 06:03:44.852265Z 192.168.15.4 69.25.94.22 HTTP 361 GET /images/bridge_small.gif HTTP/1.1
83601 2008-07-22 06:04:24.311700Z 192.168.15.4 69.25.94.22 HTTP 719 POST /secure/submit HTTP/1.1 (applicat
83614 2008-07-22 06:04:24.564165Z 192.168.15.4 69.25.94.22 HTTP 461 GET /secure/success HTTP/1.1
83654 2008-07-22 06:04:24.902157Z 192.168.15.4 69.25.94.22 HTTP 360 GET /images/bk-message.gif HTTP/1.1
Destination: HonHaiPrecis_2e:4f:60 (00:1d:d9:2e:4f:60)
.... ..0. .... = LG bit: Globally unique addr
....0 .... = IG bit: Individual address
Source: Apple_e2:c0:ce (00:17:f2:e2:c0:ce)
.... ..0. .... = LG bit: Globally unique addr
....0 .... = IG bit: Individual address
Type: IPv4 (0x0800)
[Stream index: 19]
Frame check sequence: 0x59d1a443 [unverified]
[FCS Status: Unverified]
Internet Protocol Version 4, Src: 192.168.15.4, Dst: 69.25.94.22
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 578
Identification: 0x5bba (23482)
0000 00 1d d9 2e 4f 60 00 17 f2 e2 c0 ce 08 00 45 00
0010 02 42 5b ba 40 00 3f 06 6b 20 c0 a8 0f 04 45 19
0020 5e 16 8c 90 00 50 b4 d2 22 bc a3 e4 bb 2b 80 18
0030 fb 28 6e 7a 00 00 01 01 08 0a 26 63 1e e4 73 00
0040 2d 65 47 45 54 20 2f 73 65 63 75 72 65 2f 73 75
0050 62 6d 69 74 20 48 54 54 50 2f 31 2e 31 0d 0a 41
0060 63 63 65 70 74 3a 20 69 6d 61 67 65 2f 67 69 66
0070 2c 20 69 6d 61 67 65 2f 78 2d 78 62 69 74 6d 61
0080 70 2c 20 69 6d 61 67 65 2f 6a 70 65 67 2c 20 69
0090 6d 61 67 65 2f 70 6a 70 65 67 2c 20 61 70 70 6c
00a0 69 63 61 74 69 6f 6e 2f 78 2d 73 68 6f 63 6b 77
00b0 61 76 65 2d 66 6c 61 73 68 2c 20 2a 2f 2a 0d 0a
00c0 52 65 66 65 72 65 72 3a 20 68 74 74 70 3a 2f 2f
00d0 65 6d 61 69 6c 2e 61 62 6f 75 74 2e 63 6f 6d 2f
00e0 67 69 2f 64 79 6e 61 6d 69 63 2f 6f 66 66 73 69
00f0 74 65 2e 68 74 6d 3f 7a 69 3d 31 2f 58 4a 26 73
Specifies if this is an individual (unicast) or group (broadcast/multicast) address (eth.dst.ig), 1 bit
Packets: 94410 · Displayed: 10 (0.0%)
Profile: Default

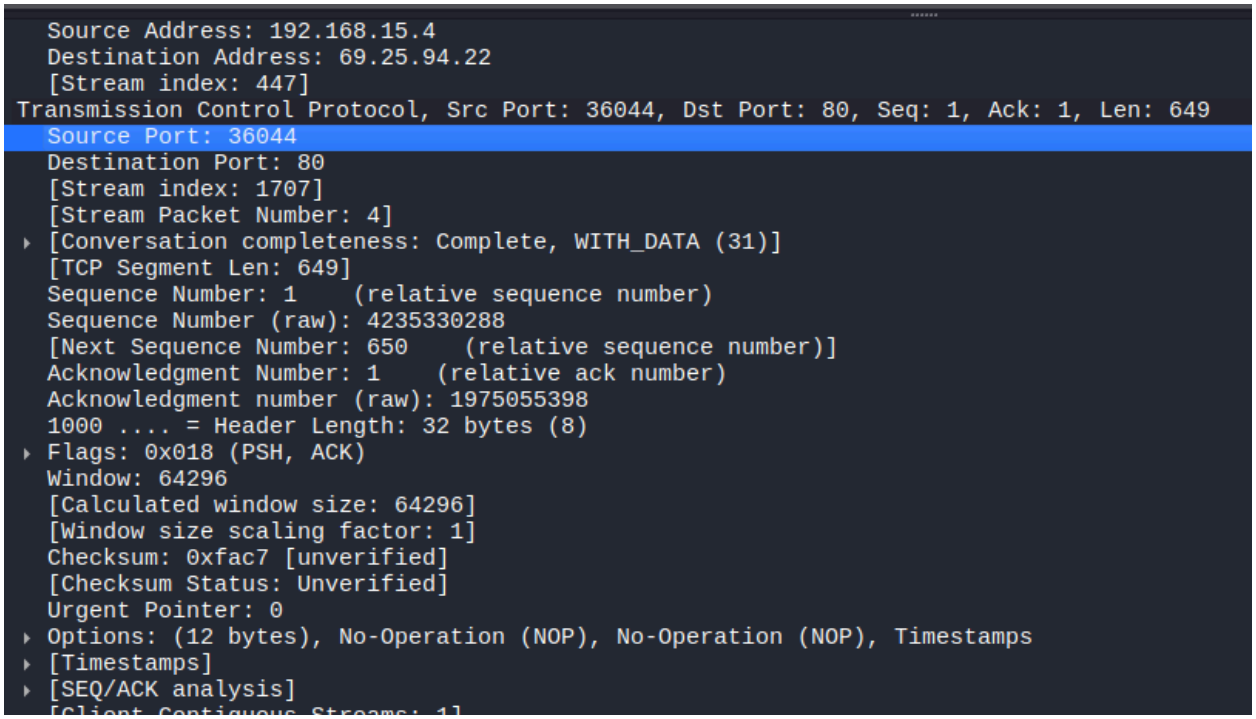

nitroba_working.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
dns.qry.name contains "willselfdestruct"
No. Time Source Destination Protocol Length Info
82912 2008-07-22 06:03:43.640466Z 192.168.15.4 192.168.1.254 DNS 88 Standard query 0x
82923 2008-07-22 06:03:43.720123Z 192.168.1.254 192.168.15.4 DNS 202 Standard query re
Frame 82912: Packet, 88 bytes on wire (704 bits), 88 bytes captured (704 bits)
Ethernet II, Src: Apple_e2:c0:ce (00:17:f2:e2:c0:ce), Dst: HonHaiPrecis_2e:4f:60 (00:1d:
Destination: HonHaiPrecis_2e:4f:60 (00:1d:d9:2e:4f:60)
.... ..0. .... = LG bit: Globally unique address (factory default)
....0 .... = IG bit: Individual address (unicast)
Source: Apple_e2:c0:ce (00:17:f2:e2:c0:ce)
.... ..0. .... = LG bit: Globally unique address (factory default)
....0 .... = IG bit: Individual address (unicast)
Type: IPv4 (0x0800)
[Stream index: 19]
Frame check sequence: 0x266fcdab [unverified]
[FCS Status: Unverified]
Internet Protocol Version 4, Src: 192.168.15.4, Dst: 192.168.1.254
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 70
Identification: 0xc25e (49758)
0000 00 1d d9 2e 4f
0010 00 46 c2 5e 00
0020 01 fe 04 01 00
0030 00 00 00 00 00
0040 65 6c 66 64 65
0050 00 01 00 01 20


Wireshark · Conversations · nitroba_working.pcap
Conversation Settings
Name resolution
Absolute start time
Display raw data
Limit to display filter
Copy
Follow Stream...
Graph...
I/O Graphs
Protocol
Bluetooth
BPv7
DCCP
DNP 3.0
Ethernet
FC
Filter list for specific type
Ethernet · 1
IPv4 · 1
IPv6
TCP · 10
UDP
Address A Address B Packets Bytes Stream ID Total Packets Percent Filtered Packets A → B Bytes A →
192.168.15.4 69.25.94.22 10 4 kB 447 207 4.83% 10 4 k


nitroba_working.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
ip.addr == 69.25.94.22 && !(ip.addr == 192.168.15.4)
No. Time Source Destination Protocol Length Info


frame contains "willselfdestruct"
No. Time Source Destination Protocol Length Info
82897 2008-07-22 06:03:43.528643Z 192.168.15.4 208.185.127.33 HTTP 746 GET /?zi=1/XJ&sdn=
82898 2008-07-22 06:03:43.549020Z 208.185.127.33 192.168.15.4 TCP 110 [TCP Previous segm
82902 2008-07-22 06:03:43.551895Z 208.185.127.33 192.168.15.4 TCP 1466 [TCP Fast Retransm
82906 2008-07-22 06:03:43.569272Z 192.168.15.4 208.185.127.40 HTTP 793 GET /gi/dynamic/of
82911 2008-07-22 06:03:43.615843Z 192.168.15.4 208.185.127.40 HTTP 1118 GET /gi/dynamic/zo
82912 2008-07-22 06:03:43.640466Z 192.168.15.4 192.168.1.254 DNS 88 Standard query 0x0
82923 2008-07-22 06:03:43.720123Z 192.168.1.254 192.168.15.4 DNS 202 Standard query res
82925 2008-07-22 06:03:43.757607Z 192.168.15.4 208.185.127.35 HTTP 842 GET /6/js/b.txt?s=
82930 2008-07-22 06:03:43.798856Z 192.168.15.4 208.185.127.35 HTTP 840 GET /6/g/email/b/b
82936 2008-07-22 06:03:43.825871Z 192.168.15.4 69.25.94.22 HTTP 596 GET /secure/submit
82939 2008-07-22 06:03:43.848398Z 192.168.15.4 208.185.127.35 HTTP 836 GET /f/lg/af1.gif
82940 2008-07-22 06:03:43.869763Z 192.168.15.4 64.236.76.160 HTTP 800 GET /rtx/r.js?cmd=
Frame 82897: Packet, 746 bytes on wire (5968 bits), 746 bytes captured (5968 bits)
Ethernet II, Src: Apple_e2:c0:ce (00:17:f2:e2:c0:ce), Dst: HonHaiPrecis_2e:4f:60 (00:1d:d
Internet Protocol Version 4, Src: 192.168.15.4, Dst: 208.185.127.33
Transmission Control Protocol, Src Port: 35962, Dst Port: 80, Seq: 791, Ack: 1005, Len: 6
Source Port: 35962
Destination Port: 80
[Stream index: 1672]
[Stream Packet Number: 7]
[Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 688]
Sequence Number: 791 (relative sequence number)
Sequence Number (raw): 3090349166
[Next Sequence Number: 1479 (relative sequence number)]
Acknowledgment Number: 1005 (relative ack number)
Acknowledgment number (raw): 273434197
0101 .... = Header Length: 20 bytes (5)
Flags: 0x018 (PSH, ACK)
Window: 64768
0030 fd 00 6a f8 00
0040 2f 58 4a 26 73
0050 6e 3d 63 6f 6d
0060 67 70 73 3d 31
0070 5f 35 31 31 26
0080 34 2e 39 2e 33
0090 31 2e 33 33 36
00a0 74 3d 31 26 62
00b0 70 25 33 41 2f
00c0 6c 66 64 65 73
00d0 65 63 75 72 65
00e0 50 2f 31 2e 31
00f0 2f 2a 0d 0a 52
0100 70 3a 2f 2f 65
0110 63 6f 6d 2f 6f
0120 65 6d 61 69 6c
0130 2f 77 69 6c 6c
0140 2e 68 74 6d 0d


ip.src == 192.168.15.4 && ip.dst == 69.25.94.22 && http.request.method == "POST"
No. Time Source Destination Protocol Length Info
83601 2008-07-22 06:04:24.311700Z 192.168.15.4 69.25.94.22 HTTP 719 POST /secure/submit HTTP/1.1 (applica
[Content length: 188]
Connection: Keep-Alive\r\n
Cache-Control: no-cache\r\n
\r\n
[Response in frame: 83604]
[Full request URI: http://www.willselfdestruct.com/secure/subm
File Data: 188 bytes
HTML Form URL Encoded: application/x-www-form-urlencoded
Form item: "to" = "lilytuckrige@yahoo.com"
Key: to
Value: lilytuckrige@yahoo.com
Form item: "from" = ""
Form item: "subject" = "you can't find us"
Form item: "message" = "and you can't hide from us.\r\n\r\nSto
Form item: "type" = "0"
Form item: "ttl" = "30"
01d0 65 6e 67 74 68 3a 20 31 38 38 0d 0a 43 6f 6e 6e ength: 1
01e0 65 63 74 69 6f 6e 3a 20 4b 65 65 70 2d 41 6c 69 ection:
01f0 76 65 0d 0a 43 61 63 68 65 2d 43 6f 6e 74 72 6f ve Cach
0200 6c 3a 20 6e 6f 2d 63 61 63 68 65 0d 0a 0d 0a 74 l: no-ca
0210 6f 3d 6c 69 6c 79 74 75 63 6b 72 69 67 65 40 79 o=lilytu
0220 61 68 6f 6f 2e 63 6f 6d 26 66 72 6f 6d 3d 26 73 ahoo.com
0230 75 62 6a 65 63 74 3d 79 6f 75 2b 63 61 6e 25 32 ubject=y
0240 37 74 2b 66 69 6e 64 2b 75 73 26 6d 65 73 73 61 7t+find+
0250 67 65 3d 61 6e 64 2b 79 6f 75 2b 63 61 6e 25 32 ge=and+y
0260 37 74 2b 68 69 64 65 2b 66 72 6f 6d 2b 75 73 2e 7t+hide+
0270 25 30 44 25 30 41 25 30 44 25 30 41 53 74 6f 70 %0D%0A%0
0280 2b 74 65 61 63 68 69 6e 67 2e 25 30 44 25 30 41 +teachin
0290 25 30 44 25 30 41 53 74 61 72 74 2b 72 75 6e 6e %0D%0ASt
02a0 69 6e 67 2e 2b 26 74 79 70 65 3d 30 26 74 74 6c ing.+&ty
02b0 3d 33 30 26 73 75 62 6d 69 74 2e 78 3d 39 32 26 =30&subm
02c0 73 75 62 6d 69 74 2e 79 3d 32 36 60 c8 d0 a3 submit.y


Source Address: 192.168.15.4
Destination Address: 69.25.94.22
[Stream index: 447]
Transmission Control Protocol, Src Port: 36044, Dst Port: 80, Seq: 1, Ack: 1, Len: 649
Source Port: 36044
Destination Port: 80
[Stream index: 1707]
[Stream Packet Number: 4]
[Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 649]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 4235330288
[Next Sequence Number: 650 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 1975055398
1000 .... = Header Length: 32 bytes (8)
Flags: 0x018 (PSH, ACK)
Window: 64296
[Calculated window size: 64296]
[Window size scaling factor: 1]
Checksum: 0xfac7 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
[Timestamps]
[SEQ/ACK analysis]

Wireshark · Follow TCP Stream (tcp.stream eq 1707) · nitroba_working.pcap

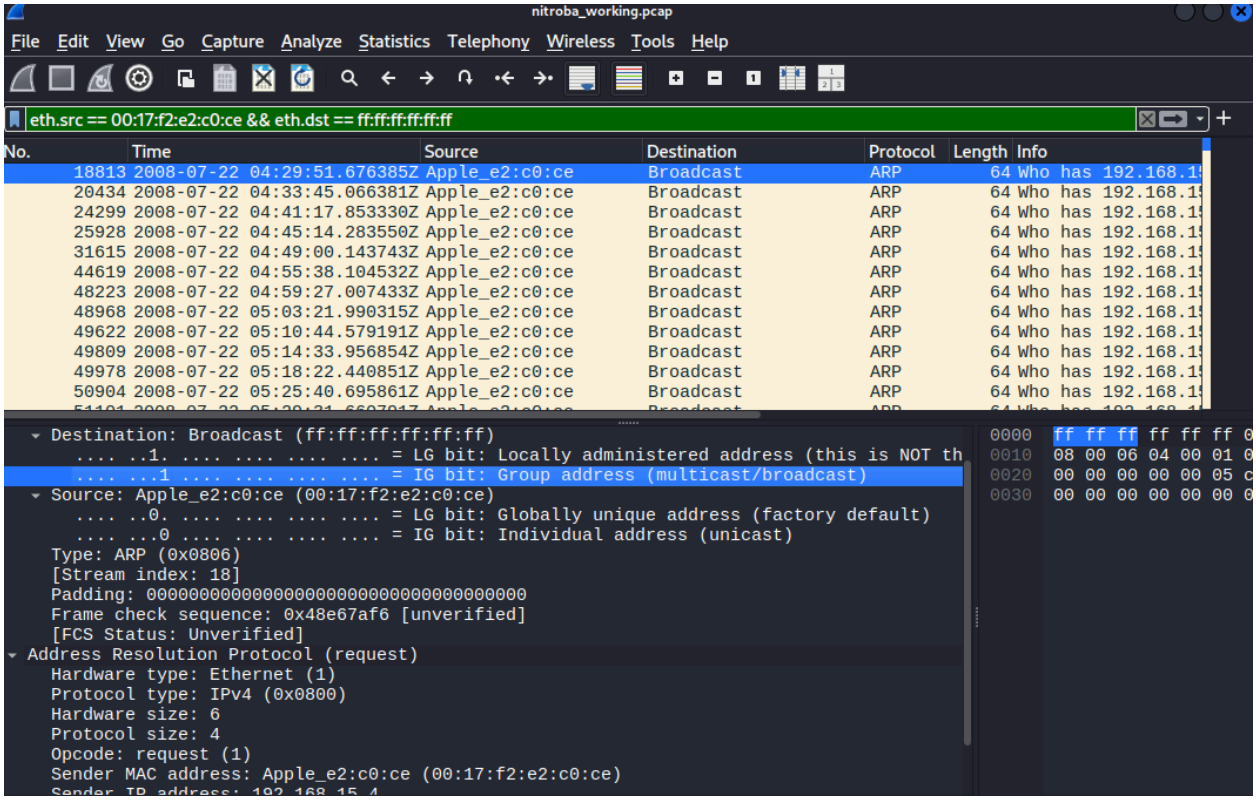
```
POST /secure/submit HTTP/1.1
Accept: image/gif, image/x-xbitmap, image/jpeg, image/pjpeg, application/x-shockwave-flash,
*/*
Referer: http://www.willselfdestruct.com/secure/submit
Accept-Language: en-us
Content-Type: application/x-www-form-urlencoded
Accept-Encoding: gzip, deflate
User-Agent: Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; SV1)
Host: www.willselfdestruct.com
Content-Length: 188
Connection: Keep-Alive
Cache-Control: no-cache

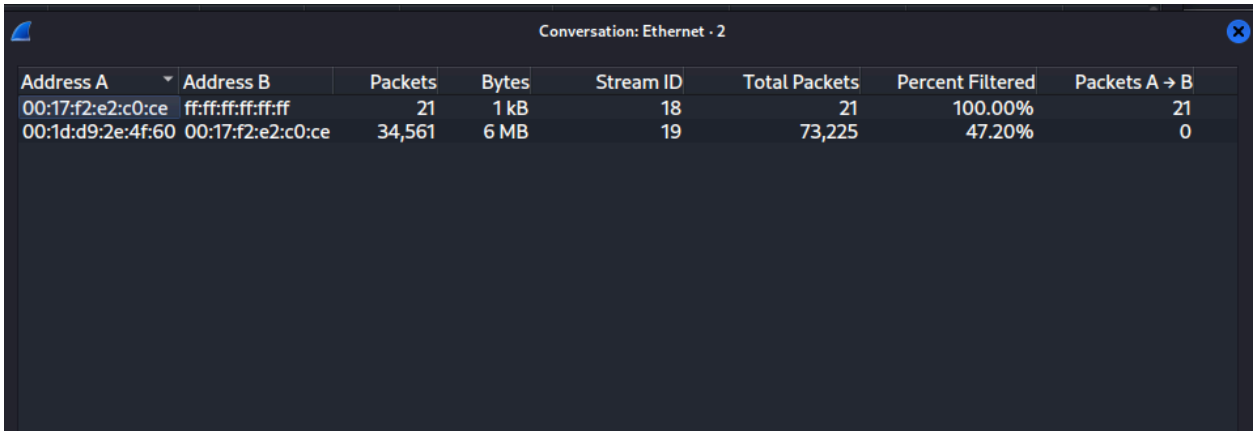
to=lilytuckrige@yahoo.com&from=&subject=you+can%27t+find+us&message=and+you+can%27t+hide+fr
om+us.%0D%0A%0D%0AStop+teaching.%0D%0A%0D%0AStart+running.+&type=0&ttl=30&submit.x=92&submi
t.y=26
HTTP/1.1 302 Moved Temporarily
Date: Tue, 22 Jul 2008 07:24:45 GMT
Server: Apache
Set-Cookie: JSESSIONID=C16BFAC8710DA82AB3D8D80AE0F4F05A; Path=/
Location: http://www.willselfdestruct.com/secure/success
Content-Length: 0
Connection: close
Content-Type: text/plain; charset=UTF-8
```

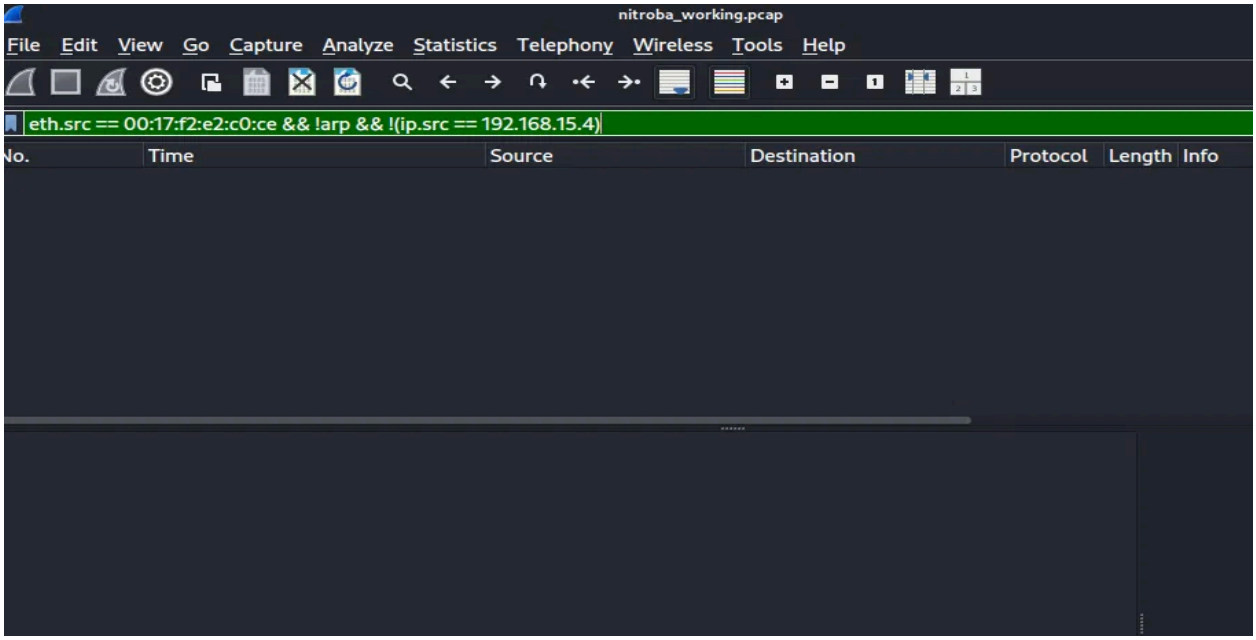
No.	Time	Source	Destination	Protocol	Length	Info
83598	2008-07-22 06:04:24.301080Z	69.25.94.22	192.168.15.4	TCP	78	80 → 36044 [SYN, ACK] Seq=0 Ack=
83599	2008-07-22 06:04:24.305823Z	192.168.15.4	69.25.94.22	TCP	70	36044 → 80 [ACK] Seq=1 Ack=1 Wi
83600	2008-07-22 06:04:24.311077Z	192.168.15.4	69.22.167.247	TCP	70	35956 → 80 [FIN, ACK] Seq=365 A
83601	2008-07-22 06:04:24.311700Z	192.168.15.4	69.25.94.22	HTTP	719	POST /secure/submit HTTP/1.1
83602	2008-07-22 06:04:24.318712Z	69.22.167.247	192.168.15.4	TCP	70	80 → 35956 [FIN, ACK] Seq=14919
83603	2008-07-22 06:04:24.406734Z	69.25.94.22	192.168.15.4	TCP	70	80 → 36044 [ACK] Seq=1 Ack=650
83604	2008-07-22 06:04:24.469169Z	69.25.94.22	192.168.15.4	HTTP	359	HTTP/1.1 302 Moved Temporarily
83605	2008-07-22 06:04:24.469263Z	69.25.94.22	192.168.15.4	TCP	70	80 → 36044 [FIN, ACK] Seq=290 A
83606	2008-07-22 06:04:24.471575Z	192.168.15.4	69.25.94.22	TCP	70	36044 → 80 [ACK] Seq=650 Ack=29
83607	2008-07-22 06:04:24.471964Z	192.168.15.4	69.25.94.22	TCP	70	36044 → 80 [ACK] Seq=650 Ack=29
83608	2008-07-22 06:04:24.472590Z	192.168.15.4	69.25.94.22	TCP	70	36044 → 80 [FIN, ACK] Seq=650 A
83609	2008-07-22 06:04:24.475027Z	192.168.15.4	69.25.94.22	TCP	82	36046 → 80 [SYN] Seq=0 Win=6424

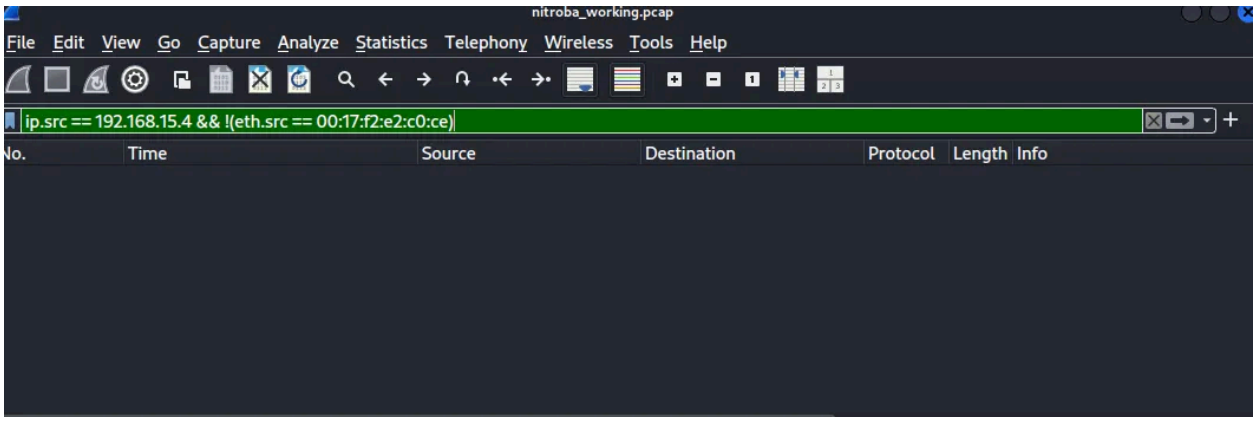
```
▸ Frame 83601: Packet, 719 bytes on wire (5752 bits), 719 bytes captured (5752 bits)
▾ Ethernet II, Src: Apple_e2:c0:ce (00:17:f2:e2:c0:ce), Dst: HonHaiPrecis_2e:4f:60 (00:1d:d9
  ▾ Destination: HonHaiPrecis_2e:4f:60 (00:1d:d9:2e:4f:60)
      .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
      .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
  ▾ Source: Apple_e2:c0:ce (00:17:f2:e2:c0:ce)
      .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
      .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Type: IPv4 (0x0800)
    [Stream index: 19]
    Frame check sequence: 0x60c8d0a3 [unverified]
    [FCS Status: Unverified]
▾ Internet Protocol Version 4, Src: 192.168.15.4, Dst: 69.25.94.22
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  ▸ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 701
    Identification: 0x02ca (714)
```

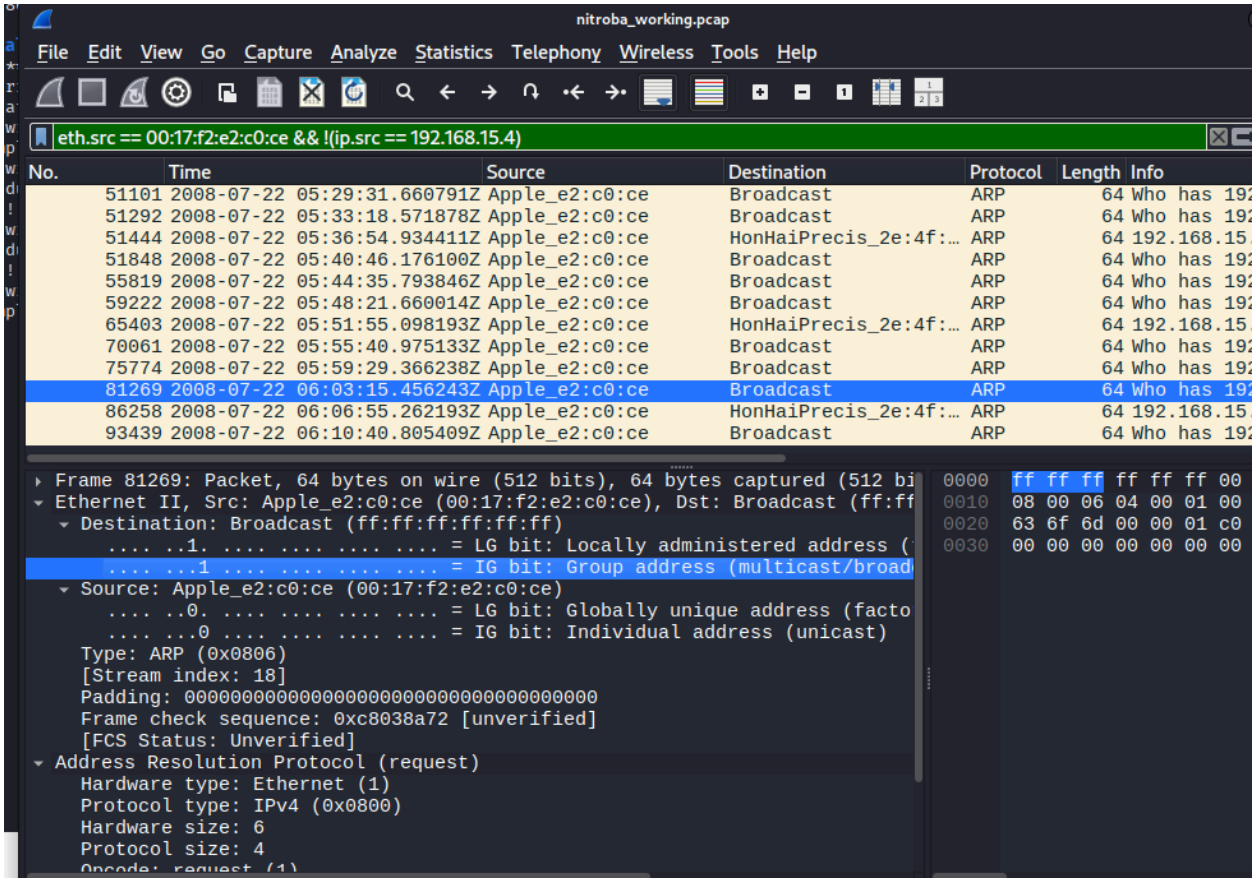
```
0020  5e 16 8c cc 00 50 fc 72  02
0030  fb 28 fa c7 00 00 01 01  08
0040  cb 95 50 4f 53 54 20 2f  73
0050  75 62 6d 69 74 20 48 54  54
0060  41 63 63 65 70 74 3a 20  69
0070  66 2c 20 69 6d 61 67 65  2f
0080  61 70 2c 20 69 6d 61 67  65
0090  69 6d 61 67 65 2f 70 6a  70
00a0  6c 69 63 61 74 69 6f 6e  2f
00b0  77 61 76 65 2d 66 6c 61  73
00c0  0a 52 65 66 65 72 65 72  3a
00d0  2f 77 77 77 2e 77 69 6c  6c
00e0  74 72 75 63 74 2e 63 6f  6d
00f0  2f 73 75 62 6d 69 74 0d  0a
0100  4c 61 6e 67 75 61 67 65  3a
0110  0a 43 6f 6e 74 65 6e 74  2d
0120  70 70 6c 69 63 61 74 69  6f
0130  2d 66 6f 72 6d 2d 75 72  6c
```

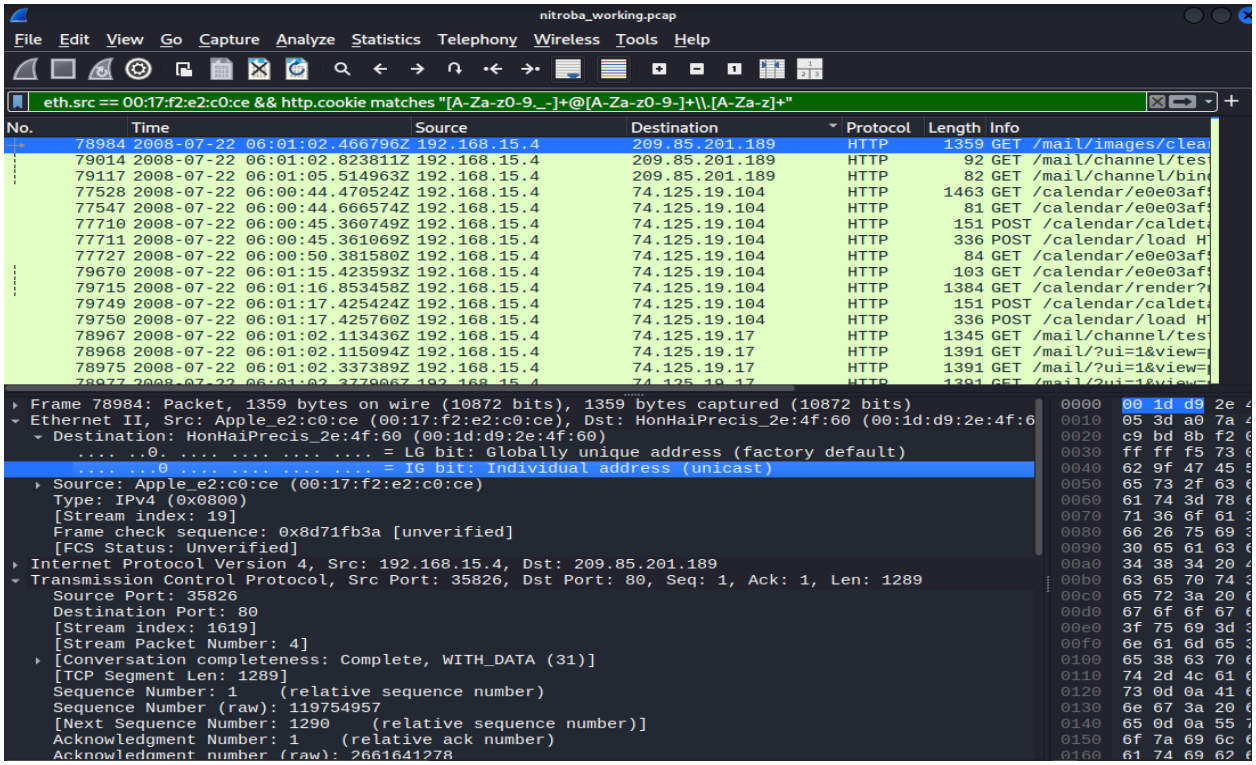

nitroba_working.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
eth.src == 00:17:f2:e2:c0:ce && eth.dst == ff:ff:ff:ff:ff:ff
No. Time Source Destination Protocol Length Info
18813 2008-07-22 04:29:51.676385Z Apple_e2:c0:ce Broadcast ARP 64 Who has 192.168.1
20434 2008-07-22 04:33:45.066381Z Apple_e2:c0:ce Broadcast ARP 64 Who has 192.168.1
24299 2008-07-22 04:41:17.853330Z Apple_e2:c0:ce Broadcast ARP 64 Who has 192.168.1
25928 2008-07-22 04:45:14.283550Z Apple_e2:c0:ce Broadcast ARP 64 Who has 192.168.1
31615 2008-07-22 04:49:00.143743Z Apple_e2:c0:ce Broadcast ARP 64 Who has 192.168.1
44619 2008-07-22 04:55:38.104532Z Apple_e2:c0:ce Broadcast ARP 64 Who has 192.168.1
48223 2008-07-22 04:59:27.007433Z Apple_e2:c0:ce Broadcast ARP 64 Who has 192.168.1
48968 2008-07-22 05:03:21.990315Z Apple_e2:c0:ce Broadcast ARP 64 Who has 192.168.1
49622 2008-07-22 05:10:44.579191Z Apple_e2:c0:ce Broadcast ARP 64 Who has 192.168.1
49809 2008-07-22 05:14:33.956854Z Apple_e2:c0:ce Broadcast ARP 64 Who has 192.168.1
49978 2008-07-22 05:18:22.440851Z Apple_e2:c0:ce Broadcast ARP 64 Who has 192.168.1
50904 2008-07-22 05:25:40.695861Z Apple_e2:c0:ce Broadcast ARP 64 Who has 192.168.1
Destination: Broadcast (ff:ff:ff:ff:ff:ff)
.... ..1. .... = LG bit: Locally administered address (this is NOT th
....1 .... = IG bit: Group address (multicast/broadcast)
Source: Apple_e2:c0:ce (00:17:f2:e2:c0:ce)
.... ..0. .... = LG bit: Globally unique address (factory default)
....0 .... = IG bit: Individual address (unicast)
Type: ARP (0x0806)
[Stream index: 18]
Padding: 000000000000000000000000000000000000
Frame check sequence: 0x48e67af6 [unverified]
[FCS Status: Unverified]
Address Resolution Protocol (request)
Hardware type: Ethernet (1)
Protocol type: IPv4 (0x0800)
Hardware size: 6
Protocol size: 4
Opcode: request (1)
Sender MAC address: Apple_e2:c0:ce (00:17:f2:e2:c0:ce)
0000 ff ff ff ff ff ff 0
0010 08 00 06 04 00 01 0
0020 00 00 00 00 00 05 c
0030 00 00 00 00 00 00 0


Conversation: Ethernet · 2
Address A Address B Packets Bytes Stream ID Total Packets Percent Filtered Packets A → B
00:17:f2:e2:c0:ce ff:ff:ff:ff:ff:ff 21 1 kB 18 21 100.00% 21
00:1d:d9:2e:4f:60 00:17:f2:e2:c0:ce 34,561 6 MB 19 73,225 47.20% 0


nitroba_working.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
eth.src == 00:17:f2:e2:c0:ce && !arp && !(ip.src == 192.168.15.4)
No. Time Source Destination Protocol Length Info


nitroba_working.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
ip.src == 192.168.15.4 && !(eth.src == 00:17:f2:e2:c0:ce)
No. Time Source Destination Protocol Length Info


nitroba_working.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
eth.src == 00:17:f2:e2:c0:ce && !(ip.src == 192.168.15.4)
No. Time Source Destination Protocol Length Info
51101 2008-07-22 05:29:31.660791Z Apple_e2:c0:ce Broadcast ARP 64 Who has 192
51292 2008-07-22 05:33:18.571878Z Apple_e2:c0:ce Broadcast ARP 64 Who has 192
51444 2008-07-22 05:36:54.934411Z Apple_e2:c0:ce HonHaiPrecis_2e:4f:... ARP 64 192.168.15
51848 2008-07-22 05:40:46.176100Z Apple_e2:c0:ce Broadcast ARP 64 Who has 192
55819 2008-07-22 05:44:35.793846Z Apple_e2:c0:ce Broadcast ARP 64 Who has 192
59222 2008-07-22 05:48:21.660014Z Apple_e2:c0:ce Broadcast ARP 64 Who has 192
65403 2008-07-22 05:51:55.098193Z Apple_e2:c0:ce HonHaiPrecis_2e:4f:... ARP 64 192.168.15
70061 2008-07-22 05:55:40.975133Z Apple_e2:c0:ce Broadcast ARP 64 Who has 192
75774 2008-07-22 05:59:29.366238Z Apple_e2:c0:ce Broadcast ARP 64 Who has 192
81269 2008-07-22 06:03:15.456243Z Apple_e2:c0:ce Broadcast ARP 64 Who has 192
86258 2008-07-22 06:06:55.262193Z Apple_e2:c0:ce HonHaiPrecis_2e:4f:... ARP 64 192.168.15
93439 2008-07-22 06:10:40.805409Z Apple_e2:c0:ce Broadcast ARP 64 Who has 192
Frame 81269: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bi
Ethernet II, Src: Apple_e2:c0:ce (00:17:f2:e2:c0:ce), Dst: Broadcast (ff:ff
Destination: Broadcast (ff:ff:ff:ff:ff:ff)
.... ..1. .... = LG bit: Locally administered address (
....1 .... = IG bit: Group address (multicast/broad
Source: Apple_e2:c0:ce (00:17:f2:e2:c0:ce)
.... ..0. .... = LG bit: Globally unique address (facto
....0 .... = IG bit: Individual address (unicast)
Type: ARP (0x0806)
[Stream index: 18]
Padding: 000000000000000000000000000000000000
Frame check sequence: 0xc8038a72 [unverified]
[FCS Status: Unverified]
Address Resolution Protocol (request)
Hardware type: Ethernet (1)
Protocol type: IPv4 (0x0800)
Hardware size: 6
Protocol size: 4
0000 ff ff ff ff ff ff 00
0010 08 00 06 04 00 01 00
0020 63 6f 6d 00 00 01 c0
0030 00 00 00 00 00 00 00


nitroba_working.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
eth.src == 00:17:f2:e2:c0:ce && http.cookie matches "[A-Za-z0-9._-]+@[A-Za-z0-9-]+\\.[A-Za-z]+"
No. Time Source Destination Protocol Length Info
78984 2008-07-22 06:01:02.466796Z 192.168.15.4 209.85.201.189 HTTP 1359 GET /mail/images/clea
79014 2008-07-22 06:01:02.823811Z 192.168.15.4 209.85.201.189 HTTP 92 GET /mail/channel/tes
79117 2008-07-22 06:01:05.514963Z 192.168.15.4 209.85.201.189 HTTP 82 GET /mail/channel/bin
77528 2008-07-22 06:00:44.470524Z 192.168.15.4 74.125.19.104 HTTP 1463 GET /calendar/e0e03af
77547 2008-07-22 06:00:44.666574Z 192.168.15.4 74.125.19.104 HTTP 81 GET /calendar/e0e03af
77710 2008-07-22 06:00:45.360749Z 192.168.15.4 74.125.19.104 HTTP 151 POST /calendar/caldet
77711 2008-07-22 06:00:45.361069Z 192.168.15.4 74.125.19.104 HTTP 336 POST /calendar/load H
77727 2008-07-22 06:00:50.381580Z 192.168.15.4 74.125.19.104 HTTP 84 GET /calendar/e0e03af
79670 2008-07-22 06:01:15.423593Z 192.168.15.4 74.125.19.104 HTTP 103 GET /calendar/e0e03af
79715 2008-07-22 06:01:16.853458Z 192.168.15.4 74.125.19.104 HTTP 1384 GET /calendar/render?
79749 2008-07-22 06:01:17.425424Z 192.168.15.4 74.125.19.104 HTTP 151 POST /calendar/caldet
79750 2008-07-22 06:01:17.425760Z 192.168.15.4 74.125.19.104 HTTP 336 POST /calendar/load H
78967 2008-07-22 06:01:02.113436Z 192.168.15.4 74.125.19.17 HTTP 1345 GET /mail/channel/tes
78968 2008-07-22 06:01:02.115094Z 192.168.15.4 74.125.19.17 HTTP 1391 GET /mail/?ui=1&view=
78975 2008-07-22 06:01:02.337389Z 192.168.15.4 74.125.19.17 HTTP 1391 GET /mail/?ui=1&view=
Frame 78984: Packet, 1359 bytes on wire (10872 bits), 1359 bytes captured (10872 bits)
Ethernet II, Src: Apple_e2:c0:ce (00:17:f2:e2:c0:ce), Dst: HonHaiPrecis_2e:4f:60 (00:1d:d9:2e:4f:6
Destination: HonHaiPrecis_2e:4f:60 (00:1d:d9:2e:4f:60)
.... ..0. .... = LG bit: Globally unique address (factory default)
....0 .... = IG bit: Individual address (unicast)
Source: Apple_e2:c0:ce (00:17:f2:e2:c0:ce)
Type: IPv4 (0x0800)
[Stream index: 19]
Frame check sequence: 0x8d71fb3a [unverified]
[FCS Status: Unverified]
Internet Protocol Version 4, Src: 192.168.15.4, Dst: 209.85.201.189
Transmission Control Protocol, Src Port: 35826, Dst Port: 80, Seq: 1, Ack: 1, Len: 1289
Source Port: 35826
Destination Port: 80
[Stream index: 1619]
[Stream Packet Number: 4]
[Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 1289]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 119754957
[Next Sequence Number: 1290 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 2661641278
0000 00 1d d9 2e
0010 05 3d a0 7a
0020 c9 bd 8b f2
0030 ff ff f5 73
0040 62 9f 47 45
0050 65 73 2f 63
0060 61 74 3d 78
0070 71 36 6f 61
0080 66 26 75 69
0090 30 65 61 63
00a0 34 38 34 20
00b0 63 65 70 74
00c0 65 72 3a 20
00d0 67 6f 6f 67
00e0 3f 75 69 3d
00f0 6e 61 6d 65
0100 65 38 63 70
0110 74 2d 4c 61
0120 73 0d 0a 41
0130 6e 67 3a 20
0140 65 0d 0a 55
0150 6f 7a 69 6c
0160 61 74 69 62

```
┌──(kali㉿kali)-[~/Nitroba-Forensics]
└─$ cd ~/Nitroba-Forensics
tshark -r nitroba_working.pcap -Y 'frame.number == 78984' -T fields -e http.cookie | tr ';' '\n'
 grep -E '[A-Za-z0-9._-]+@'
gmailchat=jcoachj@gmail.com/475090

┌──(kali㉿kali)-[~/Nitroba-Forensics]
└─$ tshark -r nitroba_working.pcap -Y 'frame.number == 78984' -T fields -e http.cookie | tr ';' '
\n' | cut -d= -f1
GX
S
GMAIL_AT
GMAIL_SU
gmailchat
PREF
NID
__utmx
__utmxx
SID
TZ
GMAIL_HELP
```

```
┌──(kali㉿kali)-[~/Nitroba-Forensics]
└─$ cd ~/Nitroba-Forensics
for f in 'eth.src == 00:17:f2:e2:c0:ce && http.cookie contains "jcoachj"' \
         'http.cookie contains "jcoachj" && !(eth.src == 00:17:f2:e2:c0:ce)' \
         'eth.src == 00:17:f2:e2:c0:ce && http.user_agent contains "MSIE"' \
         'eth.src == 00:17:f2:e2:c0:ce && http.user_agent contains "Macintosh"'; do
  echo "=== $f"
  tshark -r nitroba_working.pcap -Y "$f" -t ud -T fields -e frame.number -e frame.time_epoch > /t
mp/out.txt
  echo "count: $(wc -l < /tmp/out.txt)"
  head -1 /tmp/out.txt; tail -1 /tmp/out.txt
done
=== eth.src == 00:17:f2:e2:c0:ce && http.cookie contains "jcoachj"
count: 109
77528   1216706444.470524000
84201   1216706695.380722000
=== http.cookie contains "jcoachj" && !(eth.src == 00:17:f2:e2:c0:ce)
count: 0
=== eth.src == 00:17:f2:e2:c0:ce && http.user_agent contains "MSIE"
count: 478
48627   1216702925.701515000
90504   1216706999.850757000
=== eth.src == 00:17:f2:e2:c0:ce && http.user_agent contains "Macintosh"
count: 3915
18823   1216700993.812853000
94263   1216707095.320531000
```